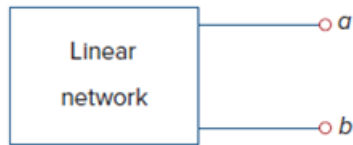


Problem

The Thevenin equivalent at terminals a - b of the linear network shown in Fig. 4.142 is to be determined by measurement. When a $10\text{-k}\Omega$ resistor is connected to terminals a - b , the voltage V_{ab} is measured as 20 V . When a $30\text{-k}\Omega$ resistor is connected to the terminals, V_{ab} is measured as 40 V . Determine: (a) the Thevenin equivalent at terminals a - b , (b) V_{ab} when a $20\text{-k}\Omega$ resistor is connected to terminals a - b .

Figure 4.142:



Step-by-step solution

Step 1 of 8

Refer to Figure 4.142 in the text book.

Replace the linear network with the Thevenin's equivalent circuit including the load resistance circuit as shown in Figure 1.

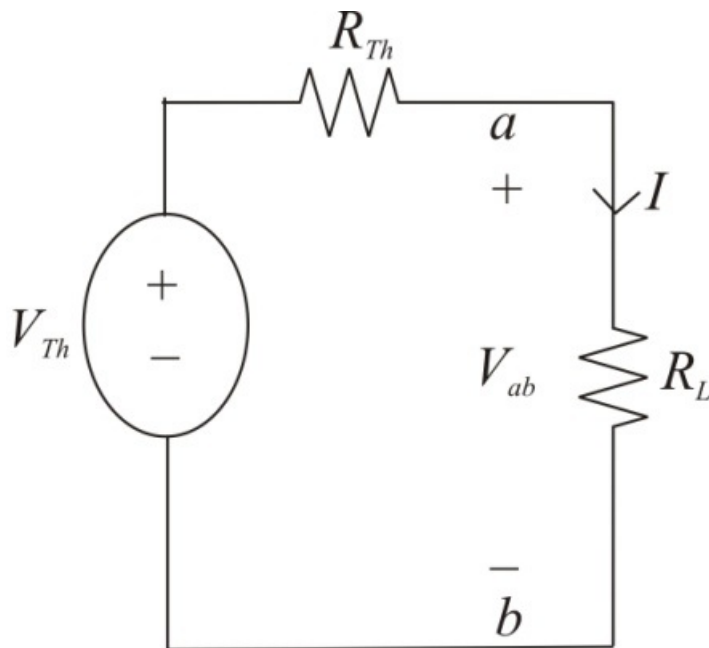


Figure 1

Step 2 of 8

Consider the expression for the current I .

$$I = \frac{V_{ab}}{R_L}$$

Consider the expression for the Thevenin's voltage V_{Th} .

$$V_{Th} = IR_{Th} + V_{ab}$$

Substitute $\frac{V_{ab}}{R_L}$ for I .

$$V_{Th} = \left(\frac{V_{ab}}{R_L} \right) R_{Th} + V_{ab}$$

$$V_{Th} = V_{ab} \left(\frac{R_{Th}}{R_L} + 1 \right) \dots\dots (1)$$

Therefore, the expression for the Thevenin's voltage V_{Th} is, $V_{ab} \left(\frac{R_{Th}}{R_L} + 1 \right)$.

Step 3 of 8

Recall equation (1).

$$V_{Th} = V_{ab} \left(\frac{R_{Th}}{R_L} + 1 \right)$$

Substitute 20 V for V_{ab} and 10 k Ω for R_L .

$$V_{Th} = (20) \left(\frac{R_{Th}}{10 \text{ k}} + 1 \right) \dots\dots (2)$$

Step 4 of 8

Recall equation (1).

$$V_{Th} = V_{ab} \left(\frac{R_{Th}}{R_L} + 1 \right)$$

Substitute 40 V for V_{ab} and 30 k Ω for R_L .

$$V_{Th} = (40) \left(\frac{R_{Th}}{30 \text{ k}} + 1 \right)$$

Substitute $(20) \left(\frac{R_{Th}}{10 \text{ k}} + 1 \right)$ for V_{Th} .

$$(20) \left(\frac{R_{Th}}{10 \text{ k}} + 1 \right) = (40) \left(\frac{R_{Th}}{30 \text{ k}} + 1 \right)$$

$$\frac{R_{Th}}{10 \text{ k}} + 1 = (2) \left(\frac{R_{Th}}{30 \text{ k}} + 1 \right)$$

$$\frac{R_{Th}}{10 \text{ k}} + 1 = \frac{R_{Th}}{15 \text{ k}} + 2$$

$$\frac{R_{Th}}{10 \text{ k}} - \frac{R_{Th}}{15 \text{ k}} = 2 - 1$$

$$\frac{3R_{Th} - 2R_{Th}}{30 \text{ k}} = 1$$

$$\frac{R_{Th}}{30 \text{ k}} = 1$$

$$R_{Th} = 30 \text{ k}\Omega$$

Therefore, the value of the Thevenin's resistance R_{Th} is 30 k Ω .

Step 5 of 8

Recall equation (2).

$$V_{Th} = (20) \left(\frac{R_{Th}}{10 \text{ k}} + 1 \right)$$

Substitute $30 \text{ k}\Omega$ for R_{Th} .

$$\begin{aligned} V_{Th} &= (20) \left(\frac{30 \text{ k}}{10 \text{ k}} + 1 \right) \\ &= (20)(3+1) \\ &= (20)(4) \\ &= 80 \text{ V} \end{aligned}$$

Therefore, the value of the Thevenins' voltage V_{Th} is 80 V .

Step 6 of 8

(a)

Draw the Thevenin's equivalent circuit diagram.

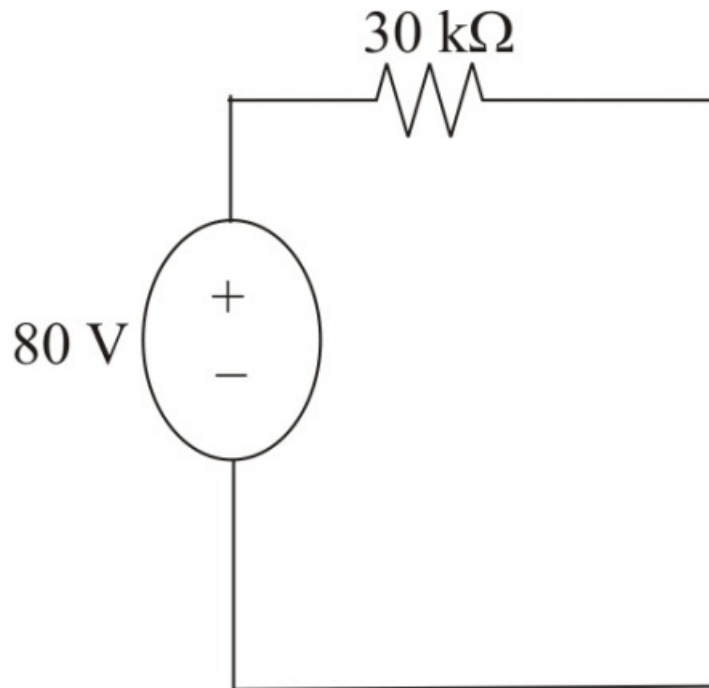


Figure 2

Step 7 of 8

(b)

Connect the load of $20\text{ k}\Omega$ to the Thevenin's circuit.

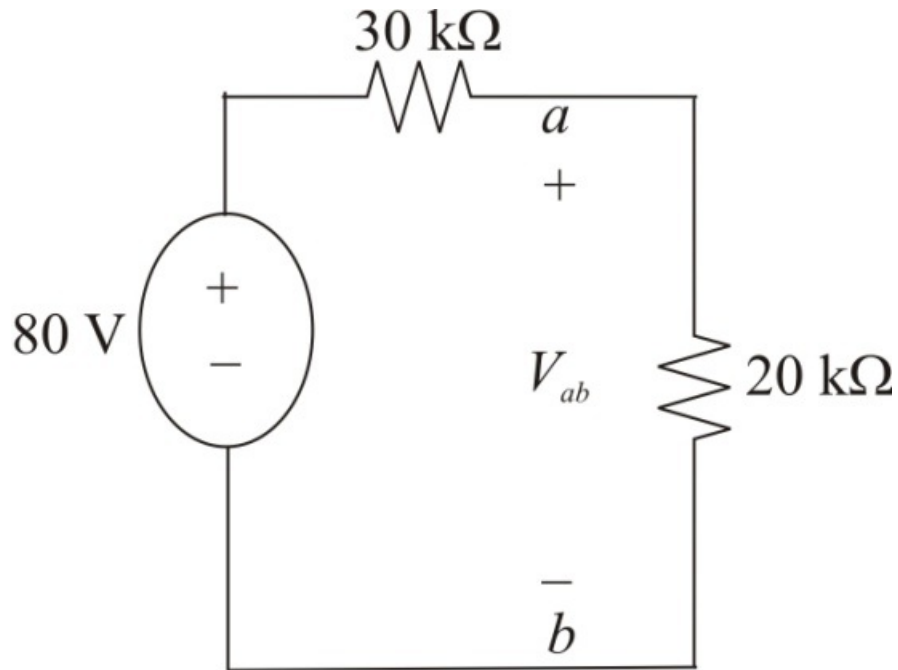


Figure 3

Step 8 of 8

Calculate the value of the voltage V_{ab} .

$$\begin{aligned} V_{ab} &= (80) \left(\frac{20\text{ k}}{20\text{ k} + 30\text{ k}} \right) \\ &= (80) \left(\frac{20\text{ k}}{50\text{ k}} \right) \\ &= (80) \left(\frac{2}{5} \right) \\ &= (16)(2) \end{aligned}$$

$$V_{ab} = 32\text{ V}$$

Therefore, the value of the voltage V_{ab} is 32 V .